VISVESWARAYA TECHNOLOGICAL UNIVERSITY

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MINI PROJECT REPORT

ON

"SMART TRASHBIN USING ULTRASONIC SENSOR"

Submitted in partial fulfilment of the requirements for the award of the degree

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

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CERTIFICATE

Certified that the mini project work entitled "SMART TRASHBIN USING ULTRASONIC SENSOR" has been successfully carried out By **Rochana U** (4UB23EC047), **Sachin GV** (4UB23EC048), **Sahana K** (4UB23EC049), **Sahanika TU** (4UB23EC050) students of University B.D.T College of Engineering in Electronics and Communication Engineering in Visvesvaraya Technological University, Belagavi during academic year 2025-26. The mini project report has been approved as it satisfies the academic respect of mini project work prescribed for the said degree.

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DECLARATION

We are the students of **5th** Semester BE,in Electronics and Communication Engineering, University B.D.T College of Engineering, Davangere hereby declare that the mini project titled "SMART TRASHBIN USING ULTRASONIC SENSOR" has been carried out by us under the guidance of **Dr. Manja Naik N** partial fulfilment of the requirements for the award of **Bachelor of Electronics & Communication Engineering** of the Visvesvaraya technological University, Belagavi during academic year 2025 -2026.

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Also, we would like to express my deepest sense of gratitude to our guide, **Dr. Manja Naik N** Professor, Department of Electronics & Communication Engineering for her constant support and guidance throughout the work.

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Project Associates

ABSTRACT

In the modern era of rapid urbanization, maintaining cleanliness and managing waste efficiently have become major environmental and social challenges. Manual handling of waste often leads to unhygienic surroundings, overflow of bins, and exposure to harmful gases. To address these issues, this project proposes the design and implementation of a Smart Dustbin System that automates waste disposal and monitoring using Arduino Uno, ultrasonic sensors, a servo motor, an LCD display, and a gas sensor.

The system operates by automatically opening the dustbin lid when an object or hand is detected near it and closing it after a short interval. Another ultrasonic sensor continuously monitors the level of waste inside the bin, while the MQ135 gas sensor detects the presence of unpleasant or toxic gases. Corresponding status messages such as *"The bin is full"* or *"Bad Odor detected"* are displayed on the LCD to alert users and maintenance personnel.

The proposed model significantly reduces human intervention, prevents overfilling, and promotes hygienic waste management practices. Its compact design, low cost, and energy efficiency make it suitable for deployment in households, public places, offices, and smart cities. This project demonstrates how embedded systems and simple automation can contribute to sustainable environmental management and cleaner living spaces.

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CHAPTER 1

INTRODUCTION

Waste management has emerged as one of the most significant challenges in modern, urbanized societies. As populations grow, so does the volume of waste generated, placing an immense strain on traditional collection systems. In many parts of India, it is a common sight to see public trash bins overflowing with garbage. This spillage creates a visual blight and establishes a breeding ground for insects, rodents, and harmful bacteria, posing a direct public health risk.

The conventional method of waste management, which relies on manual collection at fixed intervals, is increasingly inefficient. Municipal services often operate on rigid schedules, wasting fuel and manpower on routes to empty or half-full bins, while high-traffic bins overflow long before collection. The core problem is a lack of real-time information: city administrators have no way of knowing which bins are full or if a bin is creating a hygienic issue until it is manually inspected.

To address these challenges, this project proposes the design and implementation of an intelligent, sensor-based "Smart Trashbin." As a key component of "Smart City" initiatives, this project moves waste management from a reactive, schedule-based system to a proactive, data-driven one. Our system automates the waste disposal process and provides critical data about the bin's status. It is an intelligent waste bin that automatically opens its lid for hygienic, touch-free use. It detects the level of garbage and displays the bin's status on an LCD. It also detects bad odor, alerting users and maintenance staff to potential hygiene issues.

This "smart" approach is built around an Arduino UNO microcontroller and a suite of low-cost sensors. An **ultrasonic sensor** at the front detects an approaching user and triggers a **servo motor** to open the lid, creating a completely touch-free experience. An internal **level detector** (such as an IR laser or another ultrasonic sensor) constantly monitors the fill level. A **gas sensor** detects the concentration of foul odours. Finally, an **LCD screen** acts as a clear visual interface, displaying messages like "BIN FULL" or "ODOR DETECTED." This system ensures that bins are emptied only, when necessary, prevents overflow, and dramatically improves public sanitation.

1.1 Scope of Study

This mini-project focuses on the design, development, and testing of a standalone prototype of a "Smart Trashbin." The primary scope is to demonstrate the feasibility and effectiveness of using low-cost sensors and a microcontroller to create an intelligent waste management solution.

The boundaries and key functionalities to be developed within this project are:

a. Automatic Lid Operation: To successfully interface an ultrasonic sensor with a servo motor via an Arduino UNO. The system must reliably detect a user's presence (e.g., a hand) within a predefined range and trigger the lid to open automatically, then close after a short delay.
b. Waste Level Detection: To integrate a level-detection sensor (such as an IR laser or ultrasonic sensor) inside the bin. This sensor will be calibrated to measure the fill level and identify when the waste has crossed a specific threshold (e.g., 90% full).
c. Odor Detection: The gas sensor (e.g., MQ-135) is used to monitor the air quality inside the bin and detect a high concentration of foul-smelling or harmful gases.
d. User Feedback System: The LCD screen is used for output display. This screen must show the bin's status in real-time, including a welcome message, a "BIN FULL" alert when the level sensor is triggered, and an "ODOR DETECTED" alert when the gas sensor is triggered.
e. System Integration: To integrate all the above components into a single, cohesive, and self-contained prototype powered by a 9V battery or DC adapter.

1.2 Organisation of Report

This project report is structured into several chapters to present the work in a logical and comprehensive manner.

i. Chapter 1 (Introduction): This chapter provides a detailed overview of the project, discusses the problems with traditional waste management, and introduces the proposed smart trash bin solution. It also defines the scope of the project and outlines the structure of this report.

ii. Chapter 2 (Literature Survey): This chapter reviews existing research and similar projects in the field of smart waste management. It analyses different technologies and approaches used by other researchers and identifies the gaps that this project aims to address.

iii. Chapter 3 (Methodology): This chapter details the technical approach of the project. It formally presents the problem statement and lists the specific objectives. It also includes the block diagram of the proposed system, explains the working principle, and lists the hardware and software components required for implementation.

iv. Chapter 4 (System Design and Implementation): This chapter provides a detailed discussion on the hardware and software implementation. It elaborates on each component's specifications and role in the system. It also covers the assembly of the prototype and the development of the Arduino code (sketch) that controls the system's logic.

v. Chapter 5 (Results and Discussion): This chapter presents the results obtained from testing the prototype. It discusses the expected outcomes versus the actual performance and analyses the effectiveness of each subsystem (lid operation, level detection, etc.).

vi. Chapter 6 (Conclusion and Future Scope): This final chapter summarizes the project's achievements and concludes whether the objectives were met. It also discusses the applications of the smart trashbin and suggests potential enhancements and future work that can be built upon this project.

CHAPTER 2

LITERATURE SURVEY

This section highlights the research contributions in the area of smart dustbins and automated waste management. It covers the commonly used components, system architectures, and techniques explored by various authors. The literature survey helps position the current project within the context of ongoing technological advancements.

[1] SMART DUSTBIN

Authors: B.Rajapandian, K.Madhanamohan, T.Tamilselvi, R.Prithiga

(ISSN: 2249 – 8958, Volume-8, Issue-6, August 2019)

An exponential increase in Human population possesses a huge challenge to garbage management system and hence in sustaining a clean Environment. Many Cities around the World are endangered due to poor garbage management. This paper aims to find a solution by using a Smart Dustbin which is GSM and GPS enabled. It employs an 'Ultrasonic Sensor' to prevent overflow of garbage from the dustbin and a 'Gas sensor' to sense the presence of bad odour and ensures timely disposal of the unhygienic contents of the Dustbin. This paper aims at prevention of overflowing garbage from the dustbin and also ceasing unhygienic condition from prevailing near it. This paper aims to have an effective and efficient garbage disposal system.

[2] SMART DUSTBIN FOR EFFICIENT WASTE MANAGEMENT

Authors: Mahesh Chandra Bhatt1, Dharmendra Sharma2, Ashish Chauhan

(e-ISSN: 2395-0056 Volume: 06, Issue: 07 July 2019, p-ISSN: 2395-0072)

As the urbanization increases, so does the generation of waste also increases. Due to improper waste management, the garbage bins placed around are overflowed with the garbage produced on a daily basis. Also, it is not an easy task to check on a daily basis which dustbin is full. It not only makes the environment unhygienic but also is the main cause of many fatal and infectious diseases. Thus, it means that the waste management is the current necessity of the globally. In India also to promote it, the *Swachh Bharat Mission* was launched with a view of making India

clean and hygienic. This paper helps to achieve that goal with an effective method. This paper provides an effective model of an automatic dustbin for this cause. The dustbin will have sensors to detect the garbage level and once it reaches its threshold limit set, then a notification will be sent to the authorized personal to clear the waste. It will also have a sensor that will automatically open the lid of the dustbin, when a person arrives to use it, only if the dustbin is not full. This will increase its effectiveness as the pungent odour will not spread out and thus making environment more hygienic.

[3] IOT BASED SMART DUSTBIN

Authors: Telugu Maddileti , Harish Kurakula (Volume: 9, Issue: 02 February 2020, ISSN: 2277-8616)

Dustbin is used as a storage place to dispose waste, but we cannot estimate the exact amount of waste disposed by a society, and the dustbin cannot take more waste as the space should be available in it to take more. We need to know the level of waste in the dustbin and based on that we can intimate people to use the dustbin or not. In this Smart Dustbin project, we have designed a prototype where the lid of the dustbin is opened, on detection of human hand and waste, and the level of waste available inside the dustbin is sent as notification in the form of LED. The main components we used in making this prototype are Arduino, NODEMCU, Servo Motor and Ultrasonic Sensors. The software component is the application named as Blynk which is used to get notification. This dustbin can be a start to Smart Waste Management System where the officials can clean or empty the dustbin which depends on the notification received by them and not waiting for a call from a person of a society who informs the garbage trucks to come and take the waste from them.

[4] SMART TRASH BIN FOR WASTE MANAGEMENT USING ODOR SENSOR BASED ON IOT

Authors: Steffy Thankam Wilson, Tophia K. Sebastine, Merin Daniel (ISSN: 2454-132X Impact factor: 4.295 (Volume 5, Issue 2)

Increase in waste leads to many health problems and also the gases produced from wastes leads to air pollution and spreads diseases. The existing system mainly focuses on to monitor the level of the waste using the ultrasonic sensor, it monitors

the level of the waste and the notifications are sent to the workers through an android app when the bin is full. When organic waste decay H2S gas is produced, this is hazardous. In this proposed system along with the ultrasonic sensor an odor sensor, MQ136 is used to detect the H2S gas of the waste and the bin can be removed whenever there is a foul smell, even though the bin is not full. This is the main advantage of the proposed system when compared to the existing system. In this system, also there is an android app to get the notifications regarding the bin. The workers can be login to the app using Gmail. And the workers get a notification through this app if either of these conditions is true and the workers can remove the waste.

[5] SMART E-DUSTBIN

Authors: Chinmay Kolhatkar, Prachi Choudhari, Bhavesh Joshi, Dhruvin Bhuva

The prime need of a smart lifestyle begins with cleanliness and cleanliness begins with dustbin. A society will get its waste dispatched properly only if the dustbins are placed well and collected well. The main problem in the current waste management system in most of the Indian cities is the unhealthy status of dustbins. In this paper they tried to upgrade trivial and vital component of the urban waste management system, i.e. dustbin. The basic Idea behind project is to implement a smart way of handling the garbage in a smart way which is done by using the IOT protocol for transmitting the dustbin status wirelessly, which can generate email to notify to the concerned person that system is filled with garbage and need to be replaced We have selected the Espresso chip which is a node MCU ESP8266 platform. The ultrasonic sensor will show the level of garbage filled in dustbin, whereas the proximity sensors will detect the obstacle present in front of dustbin to avoid collision. LCD interfacing has been done to show the current situation of dustbin.

[6] SMART TRASH BIN

Authors: Mr.Varun Chaudhary, Mr. Rohit Kumar, Mr. Anil Rajput, Mr.Manvendra Singh, ER. Thakurendra Singh (e-ISSN: 2395-0056 Volume: 06 Issue: 05 May 2019 p-ISSN: 2395-0072)

The idea of Smart Dustbin is for the Smart buildings, Colleges, Hospitals and Bus stands. The Smart Dustbin thus thought is an improvement of normal dustbin by elevating it to be smart using sensors and logics. It will give the real time information about the level of the dustbin. It will send the message immediately when the dustbin is full. Deployment of dustbin based on actual needs. Cost of this system is minimum the resources are available easily. Improves environment quality by reducing the smell and make the cities clean. It is open automatically without touch lid. It is keep to provide environment clean. The basic operation of the system, fullness status of the bin is determined by calculating the distance between the lid of the bin and the trash by using a sensor. A distance threshold will be set according to the bin dimensions. When the distance measuring sensor indicates that the bin is full, then a microcontroller board will control a GSM module to send SMS alert, that contains bin ID and alert message, to a predefined phone number. The location of the bin is predefined by a sanitary worker who will identify the filled bin by its ID, which received by the SMS alert. The system will return to default operation when the bin is emptied by the sanitary worker.

2.1 Summary of Literature Survey

Several studies and prototypes have been developed to improve waste management through automation and sensing technologies. Researchers have implemented systems using ultrasonic sensors to measure the level of waste inside bins and to detect human presence for touchless operation. The **HC-SR04** ultrasonic sensor is preferred for its low cost, accuracy, and easy interfacing with microcontrollers like the **Arduino Uno**.

The integration of the **MQ135** gas sensor has also been widely studied for detecting harmful gases such as ammonia and carbon dioxide, which are common indicators of decomposing waste. To ensure hygienic operation, servo motors **(SG90)** are used for automatic lid movement, enabling hands-free waste disposal. Display modules such as **LCDs** or IoT dashboards are employed to alert users when the bin is full or when bad odour is detected.

Existing research highlights that automation in waste management significantly improves cleanliness, reduces human effort, and enhances public hygiene. The proposed system follows these proven concepts by combining low-cost components into an efficient, reliable, and user-friendly **Smart Trashbin** suitable for both domestic and public use.

CHAPTER 3

METHODOLOGY

The methodology section explains the systematic approach followed in designing and developing the Smart Dustbin system. It outlines the sequence of hardware integration, sensor interfacing, circuit design, and program implementation carried out during the project. This structured process ensures accurate data collection, efficient operation, and reliable system performance.

3.1 Problem statement

In most cities and institutions, waste management is still carried out through traditional methods where dustbins are manually checked at regular intervals. This process is inefficient and often leads to several problems such as overflowing bins, unhygienic surroundings, and increased manpower requirements. There is a need for an automated system that can monitor the fill level of dustbins and alert the concerned authorities in real-time. However, several challenges remain:

a. **Manual Monitoring:** Dustbins are checked manually, which is time-consuming and inefficient.
b. **Overflow Issues:** Due to irregular cleaning schedules, bins often overflow, causing odour and pollution.
c. **Health and Hygiene Problems:** Overflowing waste attracts insects and stray animals, creating unhygienic and unhealthy surroundings.
d. **Wastage of Resources:** Municipal workers or cleaning staff may visit half-empty bins, leading to unnecessary fuel and labour costs.
e. **Lack of Real-Time information:** There is no system to provide real-time status of bin fill levels to waste management authorities.

The Smart Dustbin using ultrasonic sensor project aims to overcome these issues by introducing automation and sensor-based monitoring. The system will use an ultrasonic sensor to detect the level of waste inside the bin is full. This helps to maintain cleanliness, optimize waste collection routes, and reduce manual effort.

3.2 Objectives

i. To promote cleanliness and hygiene by automating the waste collection process and reducing human contact with garbage.

ii. To design an automatic lid system that opens and closes using an ultrasonic sensor when a person approaches the dustbin.

iii. To monitor the waste level inside the bin using an ultrasonic sensor and alert users when the bin is full.

iv. To reduce manual monitoring of waste bins, saving time and effort for maintenance staff.

v. To encourage proper waste disposal habits among the public through a more convenient and hygienic system.

vi. To implement microcontroller-based automation for efficient and low-cost operation.

3.3 Proposed Methodology

The proposed methodology for the Smart Trash bin using an ultrasonic Sensor involves the integration of hardware and software components to enable automatic lid operation and waste level detection. The system is designed to enhance hygiene, reduce manual handling, and promote efficient waste management. The smart trash bin consists of an Ultrasonic Sensor, Arduino Uno Microcontroller, servo motor, Gas detector and power supply.

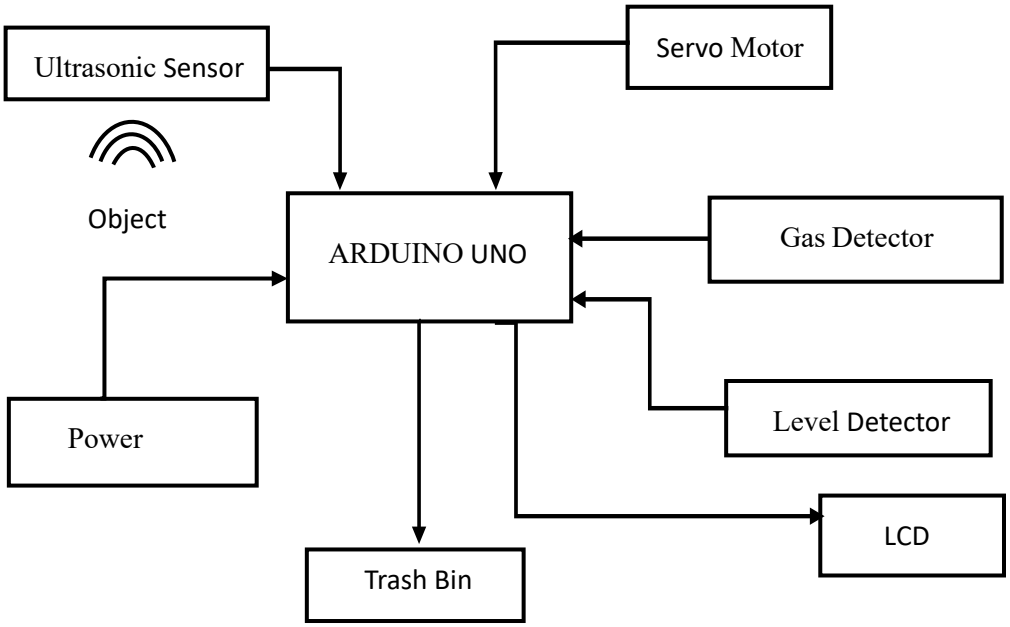


Fig 3.1: Block diagram of Smart Trashbin

The block diagram represents the functional flow of the Smart Dustbin system. The Arduino Uno acts as the central controller that coordinates all input and output components. The ultrasonic sensor detects the presence of an object or user in front of the dustbin and sends distance data to the Arduino. Based on this input, the Arduino drives the servo motor, which opens or closes the lid automatically. The level detector, implemented using another ultrasonic sensor, measures the height of the waste inside the bin to determine if it is full. The gas detector (MQ135) senses harmful or unpleasant gases emitted by the waste and provides the corresponding analog signal to the Arduino. The LCD display (16x2) shows system status messages such as “Bin is Full” or “Bad Odor Detected,” while the power supply unit provides the necessary operating voltage to all components. Together, these blocks ensure smooth operation of the smart dustbin by enabling automatic lid movement, waste level monitoring, and gas detection for hygienic waste management.

3.4 Hardware and Software requirements

Hardware Requirements

1. Arduino Uno (R3 SMD)
2. Ultrasonic Sensor (HC-SR04)
3. Servo motor (SG-90)
4. Gas Sensor (MQ135)
5. LCD Display
6. Power Supply
7. Jumper wires and Bread board

Software Requirement

Arduino IDE

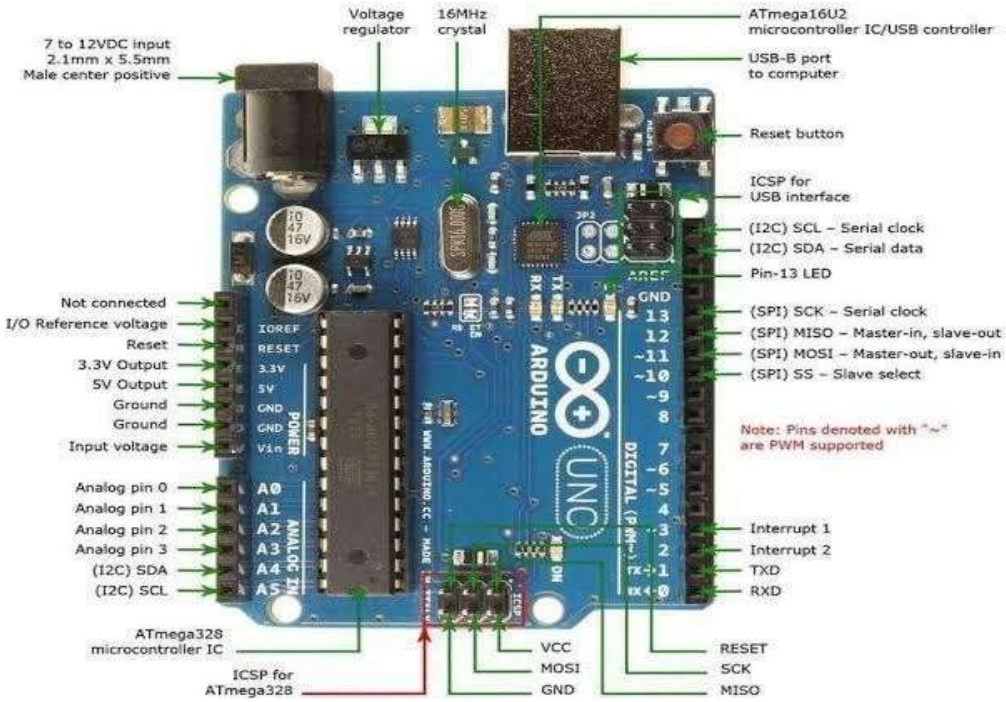
C++

CHAPTER 4

SYSTEM DESIGN AND IMPLIMENTATION

4.1 System Hardware

1.Arduino UNO R3



Uno means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform; for a comparison with previous versions, see the index of Arduino Boards.

The **Arduino Uno R3** is one of the most popular microcontroller boards in the Arduino ecosystem. Designed for beginners and experienced developers alike, it provides a versatile platform for prototyping and developing electronic projects.

Key Features

1. **Microcontroller:** ATmega328P
2. **Operating Voltage:**5V
3. **Digital I/O Pins**: 14 (of which 6 provide PWM output)
4. **Analog Input Pins**:6
5. **Input Voltage**:7-12(recommended),6-20V(limits)
6. **Clock Speed**:16MHz
7. **Flash Memory**: 32 KB (0.5 KB used by the bootloader)
8. **SRAM**: 2 KB
9. **EEPROM**: 1 KB
10. **Connectivity**: USB Type-B port for programming and power
11. **Power Pins**: 3.3V and 5V regulated outputs, GND
12. **Programming Interface**: Arduino IDE via USB connection.

Why Use Arduino Uno R3 for Smart Trash Bin Projects?

The Arduino Uno R3 is used in the smart dustbin project because it is simple, affordable, and efficient for automation applications. It controls the ultrasonic sensor, gas sensor, servo motor, and LCD effectively through its multiple input and output pins. The board operates on low power (5V) and provides stable performance for continuous operation. It can be easily programmed using the Arduino IDE through its built-in USB interface, making coding and uploading straightforward. Due to its reliability, flexibility, and wide support, the Arduino Uno R3 is an ideal microcontroller for developing a smart dustbin system.

2. Ultrasonic Sensor

The **HC-SR04 Ultrasonic Sensor** is a distance measuring device that uses ultrasonic waves to detect objects. It consists of a transmitter and receiver that operate at a frequency of 40 kHz. The transmitter emits ultrasonic pulses, which reflect back when they hit an object, and the receiver detects the echo. The sensor then calculates the distance based on the time taken for the echo to return, using the formula *Distance = (Time × Speed of Sound)/2*. It operates on a 5V power supply and provides accurate distance measurements ranging from 2 cm to 400 cm, making it suitable for obstacle detection and level monitoring applications such as the Smart Dustbin system.

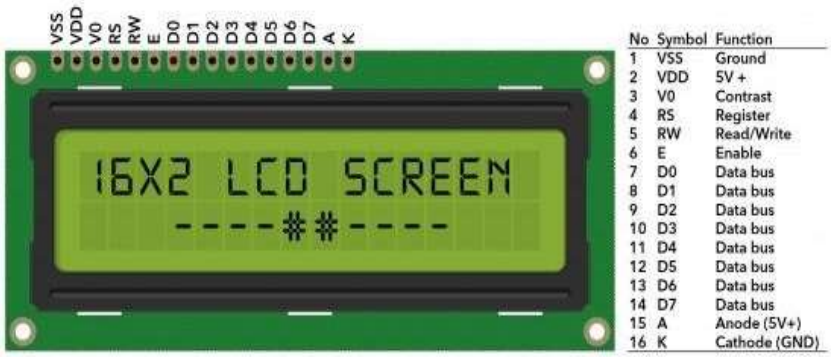
3. Servo Motor

The SG90 Servo Motor is a compact and lightweight actuator commonly used for precise angular motion control in embedded systems. It operates on a 5V DC supply and can rotate up to 180 degrees, controlled by Pulse Width Modulation (PWM) signals from the Arduino. The servo consists of a small DC motor, control circuitry, and a gear mechanism that ensures accurate positioning. It features three pins: power (VCC), ground (GND), and control (signal) making it easy to interface with microcontrollers.

4. Gas Sensor

The **MQ-135 gas sensor** is an air quality sensor designed to detect various harmful gases in the environment. It can sense gases such as ammonia (NH_3), nitrogen oxides (NO_x), benzene, carbon dioxide (CO_2), alcohol, and smoke. The sensor works by changing its electrical resistance in response to the concentration of gases in the air. This change is converted into an analog signal that can be read by a microcontroller such as the Arduino Uno.

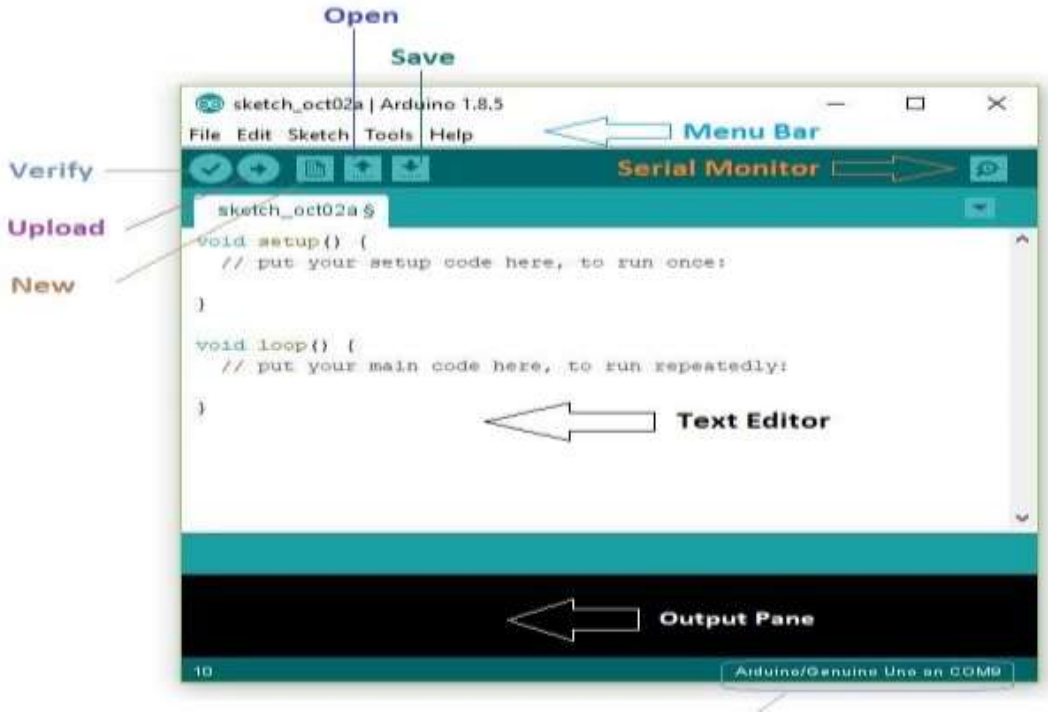
5. LCD Display



The **LCD (Liquid Crystal Display)** is an electronic display module commonly used in embedded systems and Arduino-based projects to show data in a readable form. The most widely used type is the 16x2 LCD display, which can display 16 characters per line on 2 lines. Each character is formed using a 5x8 pixel matrix, allowing letters, numbers, and symbols to be displayed clearly. In the smart dustbin project, the LCD is used to display real-time information, such as the waste level, odour detection status, or bin full alert.

4.2 System Software

ArduinoIDE



The Arduino IDE (Integrated Development Environment) is an open-source software platform used for writing, compiling, and uploading programs to Arduino microcontroller boards. It supports programming primarily in C and C++, with simplified syntax for beginners. The IDE provides a user-friendly interface that includes a text editor for writing code, a message area, a console, and buttons for common functions such as Verify, Upload, and Serial Monitor. It uses the avr-gcc compiler to convert C/C++ code into machine-readable instructions that the Arduino board can execute. The Arduino IDE enables quick development and testing of embedded applications, making it ideal for projects like the Smart Dustbin system that require sensor integration and real-time control.

4.3 System Implementation

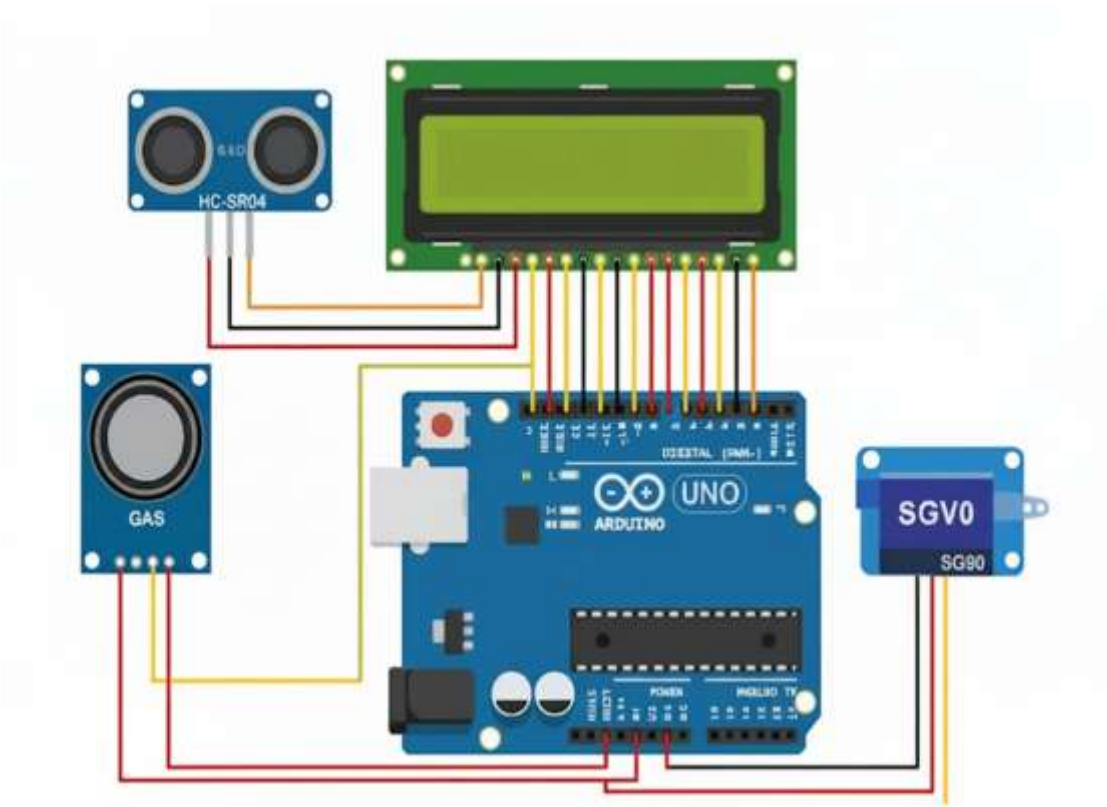


Fig4.1: Circuit Diagram

Working Principle

The working principle of the Smart Dustbin is based on real-time sensing, automatic control, and display feedback. When the system is powered ON, the Arduino Uno initializes all connected sensors and modules. The first ultrasonic sensor, placed at the top of the dustbin, continuously monitors for any object or hand in front of it. When an object is detected within a predefined distance (for example, 20 cm), the Arduino sends a control signal to the servo motor (SG90), which rotates to open the lid of the dustbin. After a short delay, the servo automatically closes the lid to maintain hygiene and save power.

Simultaneously, the second ultrasonic sensor positioned inside the bin measures the height of the accumulated waste. When the bin becomes full, the sensor output is processed by the Arduino, and a message *"The bin is full"* is displayed on the LCD screen. The system also includes a gas sensor (MQ135), which detects the presence of harmful or unpleasant gases produced by decaying waste. If the gas concentration exceeds a threshold value, the Arduino triggers an alert message *"Bad odour detected"* on the LCD.

All these operations are powered by a 9V DC power supply, and the Arduino acts as the central control unit, continuously reading sensor data, making decisions, and driving the actuators accordingly. Thus, the system ensures automatic lid operation, odour detection, and waste-level indication, making waste disposal more hygienic, convenient, and efficient.

Software: The Arduino IDE is the primary software tool used to program the Arduino UNO board. It allows users to write, compile and upload code written in **C/C++** language to the Arduino board. The program controls how the **ultrasonic sensors** and actuators respond to different inputs.

Download the Arduino Software (IDE) Download Arduino IDE from https://www.arduino.cc/en/Main/Software

Connect the board Connect your Arduino UNO hardware to PC or Laptop via USB cable.

Install the drivers

Step 1: 0pen Device Manager

Step 2: Double click the unknown Arduino Uno device; a property window pops up

Step 3: Choose the T Driver' tab, and select 'Update Driver."

Step 4: Select driver folder and click OK

Step 5: Launch the Arduino application

Step 6: Open the blink example

Step 7: Select your board

Step8: Select your serial port

Step 9: Upload the program Serial Communication Used for communication between the Arduino and a computer or other devices. Click the serial monitor button in the toolbar and select the same baud rate used in the call to begin ().

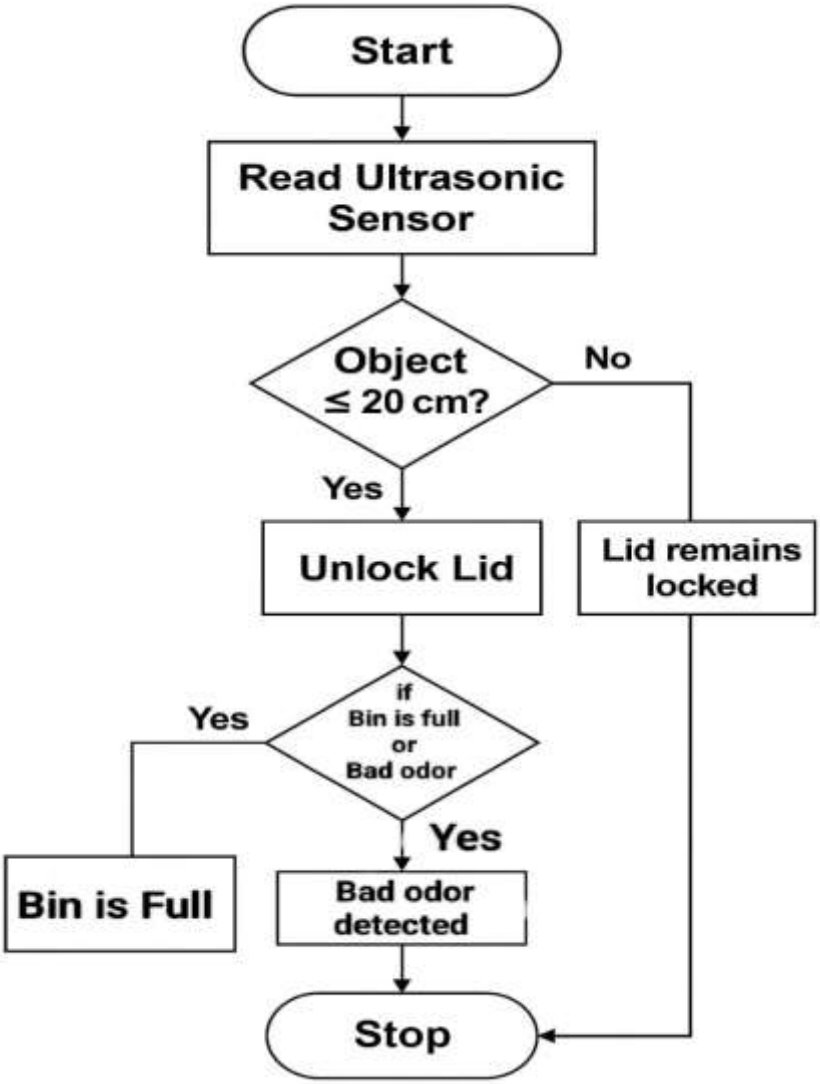


Fig: Arduino IDE Code flowchart

The flowchart describes the working process of the smart dustbin system. First, the system starts and immediately reads the distance using the ultrasonic sensor. This sensor checks if any object or person is approaching the dustbin. The program then compares the measured distance to see if the object is within 20 cm. If the object is not close, the lid remains locked and the system continues monitoring. If the object is detected within the set range, the servo motor activates and unlocks the dustbin lid. After opening the lid, the system checks the condition inside the bin. It verifies whether the bin is full using a level-detection ultrasonic sensor. The system also checks for bad odor using the MQ-135 gas sensor. If the bin is full, the system displays a “Bin is Full” alert. If bad odor is detected, it shows a “Bad Odor Detected” message before completing the cycle.

CHAPTER 5

RESULTS AND DISCUSSION

The Smart Trash bin prototype functioned effectively and met all the intended objectives. The ultrasonic sensors responded accurately by detecting both user approach and the waste level inside the bin, ensuring precise lid movement and timely “Bin Full” indication. The servo motor operated smoothly, providing quick and touchless lid opening and closing. The LCD clearly displayed system messages, allowing users to easily understand the current status of the dustbin. Overall, the system demonstrated reliable automation, improved hygiene, and successful integration of sensors and Arduino control, proving the feasibility of the smart waste management concept.

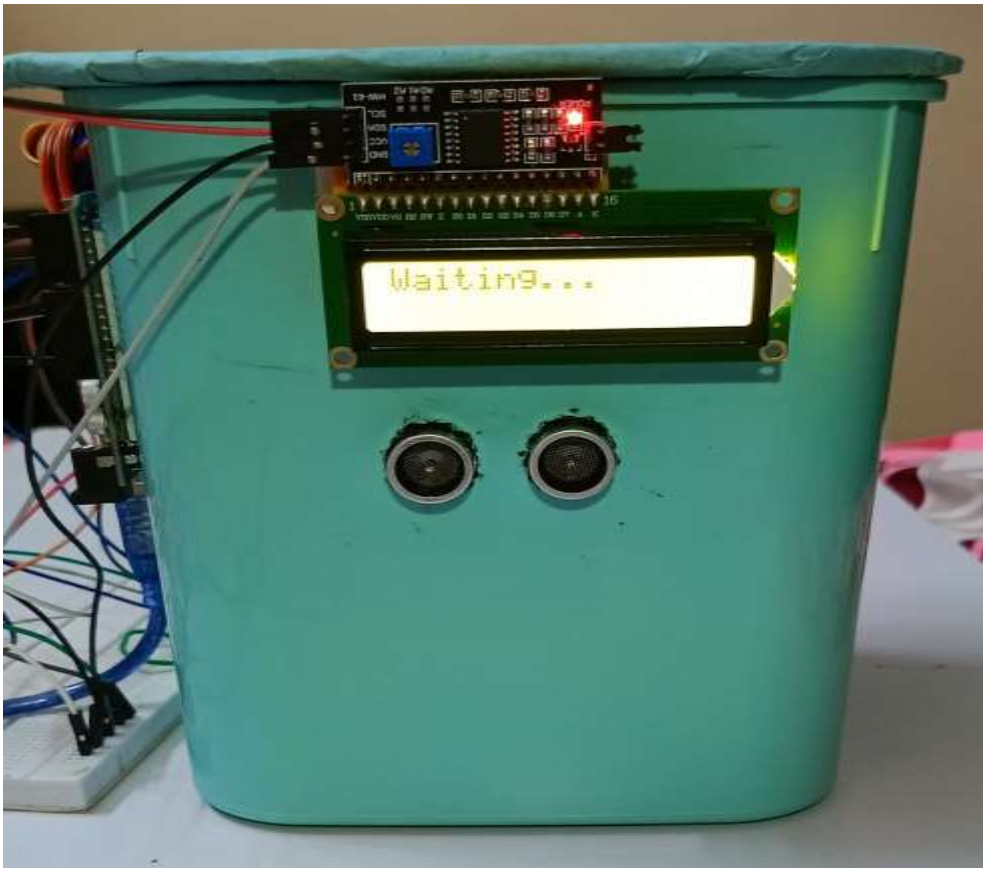


Fig 5.1: The ultrasonic sensor is waiting for object

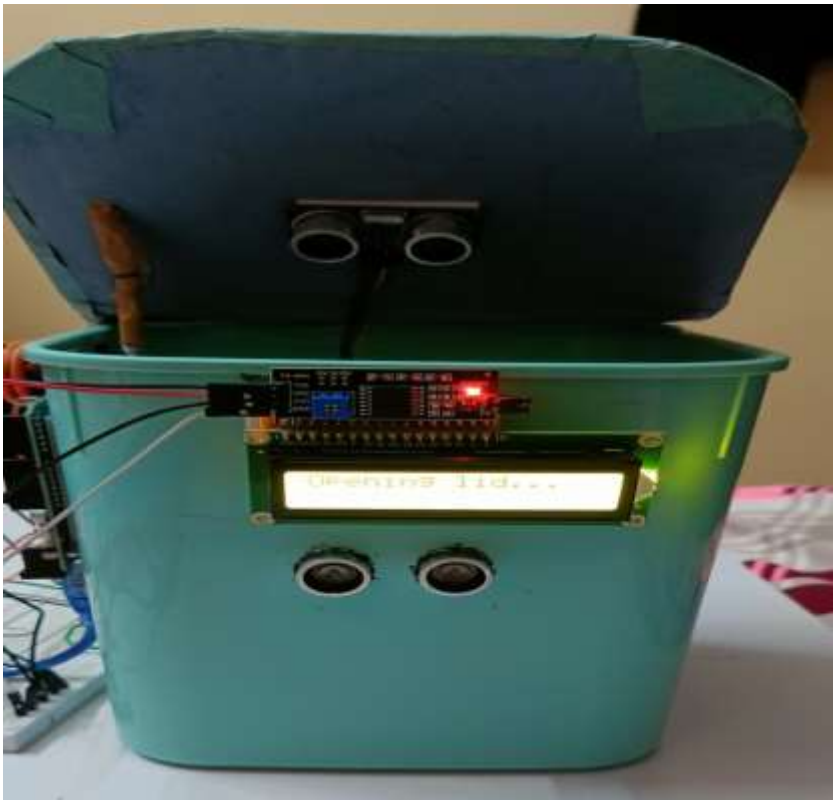


Fig 5.2: Object is detected, lid opened

Fig 5.3: Bin is full

5.2 Merits:

I. Promotes Hygiene and Cleanliness:
   - Touchless operation prevents direct contact with the bin, reducing the spread of germs and infection.
   - Minimizes unpleasant odour exposure by detecting and alerting about bad smell.

II. Efficient Waste Level Monitoring:
   - The ultrasonic level detector ensures the bin is not overfilled.
   - Alerts help timely waste collection and avoid overflow in public areas.

III. User-Friendly Feedback System:
   - LCD display provides clear messages like “Bin Full” or “Bad Odor Detected.”
   - Easy for users and maintenance staff to understand system status.

IV. Enhances Public Health & Safety:
   - Odor detection helps identify decomposing waste early, reducing exposure to harmful gases.
   - Limits contact-based transmission of germs because the lid opens automatically.

V. Reduces Manual Contact with Waste:
   - Helps maintain hygiene for users and municipal workers.
   - Prevents direct exposure to harmful gases or overfull waste.

VI. Suitable for Smart Cities:
   - Supports automation and cleanliness goals.
   - Can be implemented in homes, offices, schools, hospitals, and public places.

VII. Scalable and Upgradable:
   - System can be expanded to IoT-based monitoring.
   - More sensors (humidity, weight, temperature) can be added easily.

5.3 Limitations

I. Mechanical Wear and Tear:
   - Continuous opening and closing of the lid can cause mechanical stress on the servo motor over time.
   - Hinges or the lid mechanism may require maintenance.

II. Not Suitable for Heavy or Large Waste:
   - The automated lid may not lift properly if someone tries to throw heavy, oversized, or irregular waste items.

III. Limited Storage Capacity:
   - Even with sensors, the dustbin size is fixed, so it cannot handle large volumes during peak usage.

5.4 Scope for Future Work

I. Integration with IoT for Remote Monitoring:
   - The system can be upgraded using Wi-Fi or GSM modules to send alerts (bin full, odour detected) to mobile apps or cloud dashboards.
   - Municipal corporations can use this data to optimize waste collection routes.

II. Automatic Waste Segregation:
   - Future versions can include sensors for dry/wet waste detection or metal/non-metal identification to enable automated waste classification.

III. Mobile App Connectivity:
   - A dedicated Android/iOS app could give real-time updates, bin history, and notifications to users or maintenance staff.

IV. Enhanced Mechanical Design:
   - Adding a stronger, waterproof, and vandal-proof outer body would make the system suitable for outdoor public installations.

V. AI-Based Prediction System:
   - Machine learning can be used to predict when the bin will be full based on usage patterns, helping schedule cleaning more efficiently.

5.5 Conclusion

The Smart Trashbin system successfully demonstrates the use of sensors and automation to improve hygiene and efficiency in waste management. By integrating ultrasonic sensors, a servo motor, a gas sensor, and an LCD display with an Arduino Uno controller, the system provides touchless lid operation, real-time waste level monitoring, and odor detection. These features collectively reduce manual handling, prevent bin overflow, and promote a cleaner environment. The project highlights how simple, low-cost electronic components can be combined to create a practical and user-friendly solution suitable for homes, offices, and public spaces. Although the current prototype is non-IoT based, it lays a strong foundation for future enhancements such as mobile monitoring, solar power, and smart waste segregation. Overall, the Smart Trashbin proves to be an effective step toward modern, automated, and sustainable waste management systems.

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